Algebraic Expressions – SS2 Study Notes

These notes are designed for Senior Secondary School students in. They cover the meaning of algebraic expressions, types, operations, worked examples, and practice questions.

1. Meaning of Algebraic Expressions

An algebraic expression is a mathematical phrase made up of numbers, variables, and operations such as addition, subtraction, multiplication, and division.

Examples:
- $3x + 5$
- $2a^2 - 7a + 4$
- $\frac{5m}{n}$

In algebraic expressions:
- Variables are letters that represent numbers (x, y, a, b).
- Constants are fixed numbers (5, -3, 12).
- Coefficients are numbers multiplying variables (3 in 3x).

2. Terms in Algebraic Expressions

A term is a part of an algebraic expression separated by + or – signs.

Example:
In $4x^2 + 3x - 7$:
- $4x^2$ is a term
- $3x$ is a term
- -7 is a term

3. Types of Algebraic Expressions

1. Monomial – An expression with one term.
   Examples: $5x$, $7a^2$

2. Binomial – An expression with two terms.
   Examples: $x + 5$, $2a - 3$

3. Trinomial – An expression with three terms.
   Examples: $x^2 + 3x + 2$

4. Polynomial – An expression with many terms.
   Examples: $x^3 + 2x^2 - x + 7$

4. Like and Unlike Terms

Like terms have the same variables raised to the same powers.
Examples:
- $3x$ and $7x$ are like terms.
- $4a^2$ and $-2a^2$ are like terms.

Unlike terms have different variables or powers.
Examples:
- $3x$ and $3y$
- x and x^2

5. Simplifying Algebraic Expressions

To simplify an expression, combine like terms.

Example 1:
Simplify $3x + 5x - 2$
Solution:
$3x + 5x = 8x$
Answer $= 8x - 2$

Example 2:
Simplify $7a - 2a + 4 - 1$
Solution:
$7a - 2a = 5a$
$4 - 1 = 3$
Answer $= 5a + 3$

6. Addition and Subtraction of Algebraic Expressions

Addition Example:

$(3x + 2) + (4x - 5)$

$= 3x + 2 + 4x - 5$

$= 7x - 3$

Subtraction Example:

$(5a + 7) - (2a - 3)$

$= 5a + 7 - 2a + 3$

$= 3a + 10$

7. Multiplication of Algebraic Expressions

Example 1:

Multiply $3x$ by $4x$

$3x \times 4x = 12x^2$

Example 2:

Expand $2(x + 3)$

$= 2 \times x + 2 \times 3$

$= 2x + 6$

Example 3:

Expand $(x + 2)(x + 3)$

$= x^2 + 3x + 2x + 6$

$= x^2 + 5x + 6$

8. Factorization

Factorization is the reverse of expansion.

Example 1:

Factorize $6x + 12$

$= 6(x + 2)$

Example 2:

Factorize $x^2 + 5x + 6$
$= (x + 2)(x + 3)$

9. Substitution in Algebraic Expressions

Substitution means replacing variables with numbers.

Example:
Find the value of $2x + 3$ when $x = 4$.
Solution:
$= 2(4) + 3$
$= 8 + 3$
$= 11$

10. Common Errors Students Make

- Adding unlike terms incorrectly.
  Example: $3x + 2y \neq 5xy$

- Forgetting to change signs during subtraction.

- Incorrect expansion of brackets.

- Mixing coefficients and powers wrongly.

11. Worked Examples

Example 1:
Simplify $4x + 7 - 2x + 5$
Solution:
$4x - 2x = 2x$
$7 + 5 = 12$
Answer $= 2x + 12$

Example 2:
Expand $3(a + 4)$
Solution:
$= 3a + 12$

Example 3:
Factorize $9y - 6$
Solution:
$= 3(3y - 2)$

Example 4:
Find the value of $5m - 2$ when $m = 3$
Solution:
$= 5(3) - 2$
$= 15 - 2$
$= 13$

12. Practice Questions

A. Simplify:
1. $3x + 4x - 5$
2. $7a - 2a + 9$
3. $5y + 3 - 2y - 1$

B. Expand:
4. $2(x + 5)$
5. $4(a - 3)$
6. $(x + 2)(x + 4)$

C. Factorize:
7. $8x + 16$
8. $x^2 + 7x + 12$

D. Substitution:
9. Find $3a + 2$ when $a = 5$
10. Find $2x^2 - 1$ when $x = 3$

13. Answers to Practice Questions

1. $7x - 5$
2. $5a + 9$

3. $3y + 2$
4. $2x + 10$
5. $4a - 12$
6. $x^2 + 6x + 8$
7. $8(x + 2)$
8. $(x + 3)(x + 4)$
9. 17
10. 17

Study Tip: Practice algebra daily. Always group like terms together and work step by step.